

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

Roll No.:T003	Name: Pranay Bhadwalkar
Class:TYIT	Batch: B1
Date of Assignment:04/08/2026	Date/Time of Submission:04/08/2026

a. Develop a PWA that works in offline mode using Service Worker.

Code:

Offline.html:

```
<!DOCTYPE html>

<html>

<head>

<title>College Portal</title>

<style>

body {

font-family: Arial;

background: #f4f6f9;

margin: 0;

}

.header {

background: #003366;

color: white;

text-align: center;

padding: 10px;

}

.container {

width: 80%;

margin: auto;

margin-top: 20px;

}

.card {
```

```
background: white;
padding: 15px;
margin-top: 15px;
border-radius: 8px;
}
h3 {
color: #003366;
}
button {
padding: 8px 15px;
margin: 5px;
border: none;
color: white;
cursor: pointer;
}
.blue {
background: #007bff;
}
.green {
background: #28a745;
}
.red {
background: #dc3545;
}
.offline {
color: red;
text-align: center;
}
</style>
</head>
<body>

<div class="header">
```

```
<h2>Laxmi Charitable Trust's</h2>
```

```
<h3>Sheth L.U.J College</h3>
```

```
<p>College Portal</p>
```

```
</div>
```

```
<div class="container">
```

```
<div class="card">
```

```
<h2 style="text-align:center;">❑ You are Offline</h2>
```

```
<p class="offline">No internet connection</p>
```

```
</div>
```

```
<div class="card">
```

```
<h3>Student Details</h3>
```

```
<p>Name: Pranay Bhadwalkar</p>
```

```
<p>Roll No: T003</p>
```

```
<p>Class: TYIT</p>
```

```
<p>Batch: 01</p>
```

```
</div>
```

```
<div class="card">
```

```
<h3>College Info</h3>
```

```
<p>Department: IT</p>
```

```
<p>University: Mumbai University</p>
```

```
<p>Location: Andheri</p>
```

```
</div>
```

```
<div class="card" style="text-align:center;">
```

```
<h3>Actions</h3>
```

```
<button class="blue">Attendance</button>
```

```
<button class="green">Results</button>
```

```
<button class="blue">Assignments</button>
```

```
<button class="red">Logout</button>
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

service-worker.js:

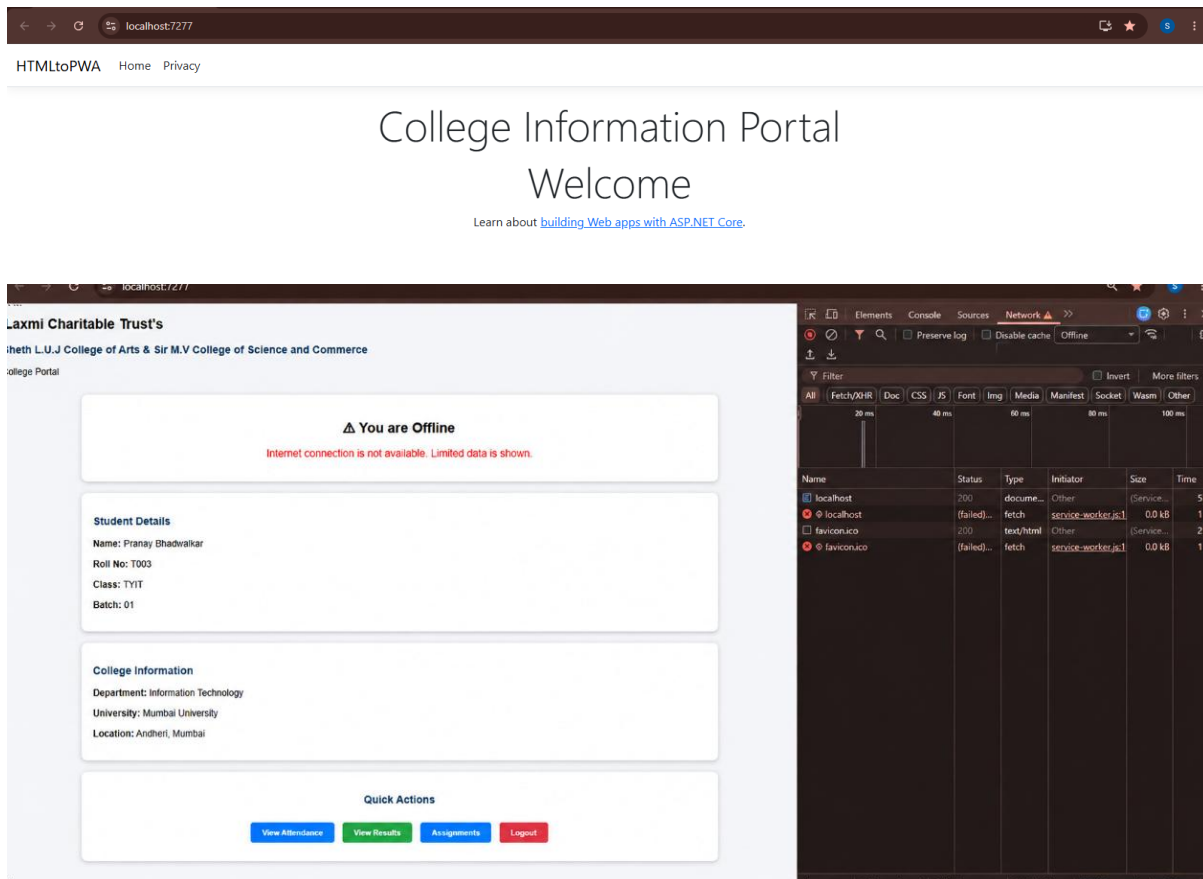
```
const CACHE = "v1";
```

```
self.addEventListener("install", event => {  
  event.waitUntil(  
    caches.open("offline-cache").then(cache => {  
      return cache.addAll([  
        "/",  
        "/offline.html"  
      ]);  
    })  
  );  
});
```

```
  console.log("Service Worker Installed");  
});
```

```
self.addEventListener("fetch", event => {  
  event.respondWith(  
    fetch(event.request).catch(() => {  
      return caches.match("/offline.html");  
    })  
  );  
});
```

Output:



b. Create a Student Record Management PWA.

Code:

HomeController.cs:

using HTMLtoPWA.Models;

using Microsoft.AspNetCore.Mvc;

using System.Diagnostics;

namespace HTMLtoPWA.Controllers

{

public class HomeController : Controller

{

public IActionResult Index()

{

Student s = new Student();

s.RollNo = 103;

```
s.Name = "Pranay";
```

```
return View(s);  
}
```

```
public IActionResult Privacy()  
{  
    return View();  
}
```

```
[ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]  
public IActionResult Error()  
{  
    return View();  
}  
}
```

Student.cs:

```
namespace HTMLtoPWA.Models  
{  
    public class Student  
    {  
        public int RollNo { get; set; }  
  
        public string Name { get; set; }  
    }  
}
```

index.cshtml:

```
@model HTMLtoPWA.Models.Student  
  
@{  
    ViewData["Title"] = "Home Page";  
}
```

```
<div class="text-center">
```

```
<h1 class="display-4">College Portal</h1>
```

```
<p>
```

```
    Learn about
```

```
    <a href="https://learn.microsoft.com/aspnet/core">
```

```
        building Web apps with ASP.NET Core
```

```
    </a>.
```

```
</p>
```

```
<h1>Offline Page</h1>
```

```
<h2>Student Record</h2>
```

```
<p>Roll No : @Model.RollNo</p>
```

```
<p>Name : @Model.Name</p>
```

```
</div>
```

service-worker.js :

```
const CACHE = "v1";
```

```
self.addEventListener("install", e => {  
    e.waitUntil(  
        caches.open(CACHE).then(cache => {  
            return cache.addAll([  
                "/",  
                "/Home/Index",  
                "/offline.html",  
                "/css/site.css",  
                "/js/site.js"
```

```
    });  
  })  
);  
});  
  
self.addEventListener("fetch", e => {  
  e.respondWith(  
    fetch(e.request).catch(() => {  
      return caches.match(e.request).then(response => {  
        return response || caches.match("/offline.html");  
      });  
    })  
  );  
});
```

Output:

College Portal

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Offline Page Student Record

Roll No : 103

Name : Pranay

[HTMLtoPWA](#) 

- [Home](#)
- [Privacy](#)

College Portal

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Offline Page

Student Record

Roll No : 103

Name : Pranay

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